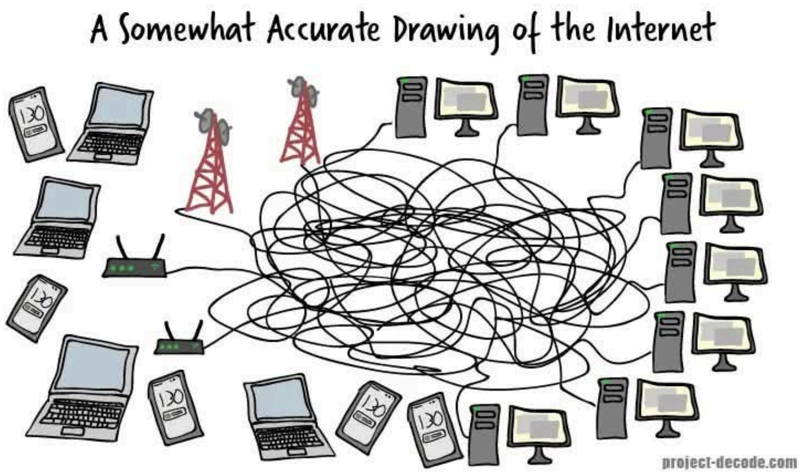
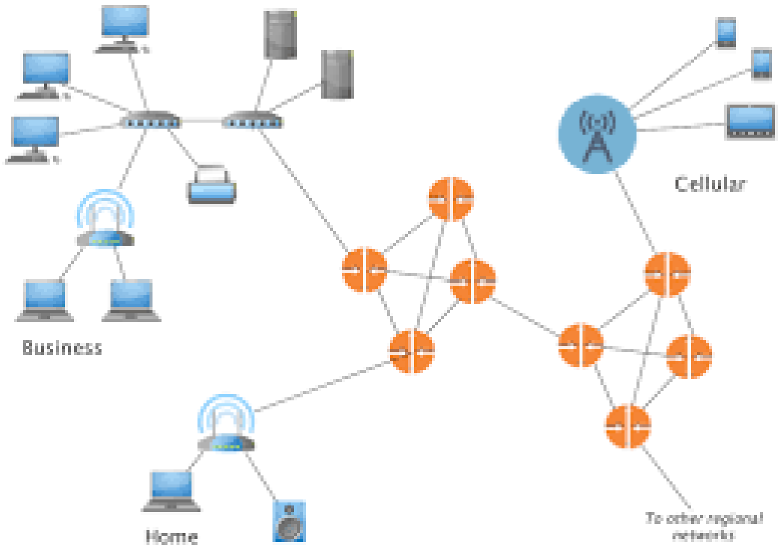
How Wireless Internet Works

With Real-World Examples

What is the Internet?

It's a network of networks connecting billions of devices worldwide using standardized protocols to communicate.





Revolutionary Idea

1. When a computer has an IP address, it can communicate with any other computer in the world that also has an IP address.
2. We can communicate regardless of the underlying hardware or physical connection.
3. This is the fundamental principle of the Internet: **interoperability** through standardized protocols.

How is this possible?

Standardization at each layer!

- Standardized protocols
- Standardized hardware interfaces

```
Application  →  HTTP, SSH, DNS
Transport    →  TCP / UDP
Network      →  IP   ← Internet
Link         →  Ethernet / Wi-Fi / LTE / Fiber
Physical     →  Cable / Radio / Light
```

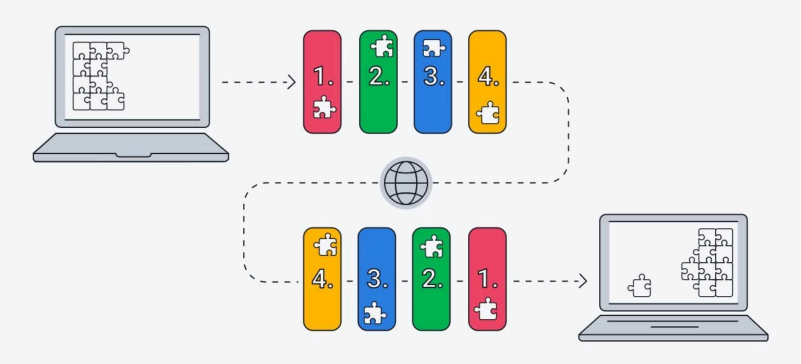
We mainly use TCP/IP, Ethernet, Cat6 Cables, and RJ45 connectors for Wired Internet

1. TCP/IP is the core protocol suite that enables communication across the Internet.
2. Ethernet (IEEE 802.3) is the most widely used link layer protocol for wired connections.
3. Cat6 cables and RJ45 connectors are the standard physical media for Ethernet connections.

```
Application
   ↑
TCP / IP
   ↑
IEEE 802.3  ← Ethernet protocol (Frame, MAC, signaling)
   ↑
Cable (Cat5e / Cat6 / Fiber)
   ↑
Connector (RJ-45 / LC / SC)
```

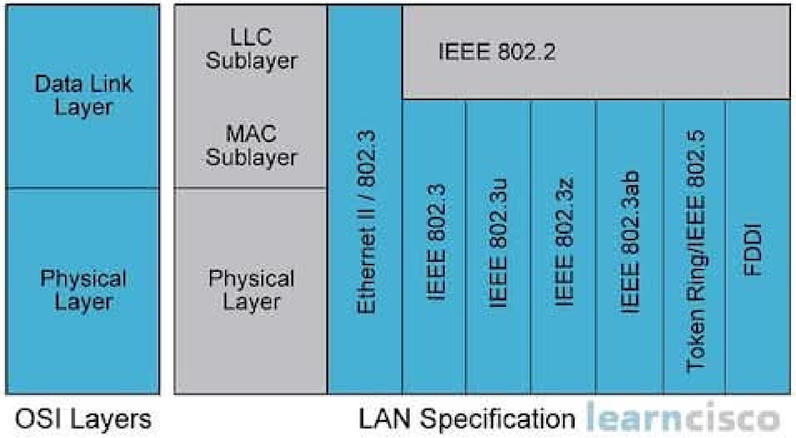
TCP/IP Protocol Suite

TCP protocol ensures reliable, ordered, and error-checked delivery of data between applications running on hosts communicating via an IP network.

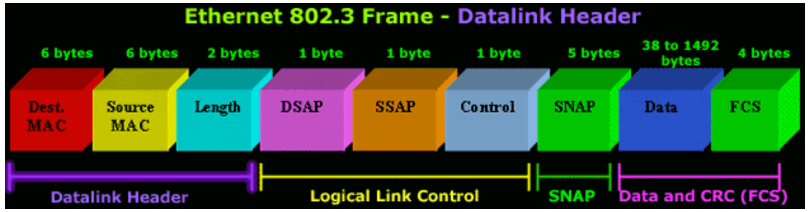


IEEE 802.3 Ethernet Standard

IEEE 802.3 define the physical (L1) and data link (L2) layers.

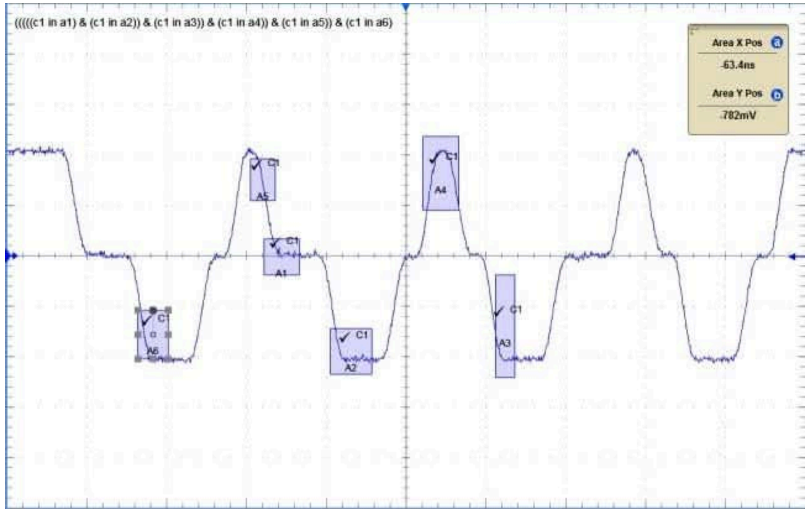


IEEE 802.3 defines the frame format, MAC addressing, and signaling for wired Ethernet connections.



IEEE 802.3 specifies different Ethernet standards based on speed and media:

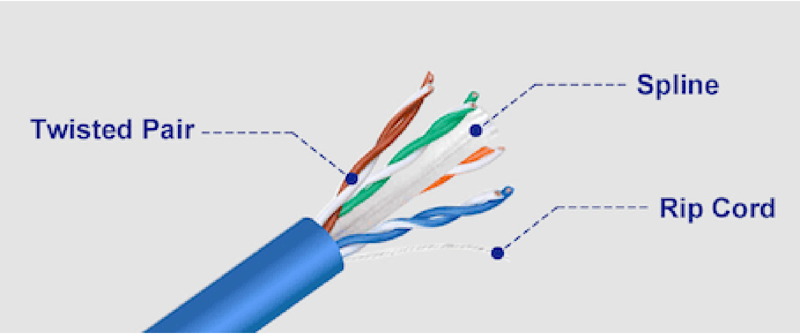
- 1000BASE-T: 1 Gbps over Cat5e/Cat6
- 10GBASE-T: 10 Gbps over Cat6a/Cat7
- 100GBASE-SR4: 100 Gbps over multimode fiber



RJ45 Connectors and Cat5e/Cat6 Cables

- RJ45 connector: 8 pins for data transmission

- Cat5e/Cat6 cables: Twisted pairs to reduce interference



How to Go Wireless?

No change in TCP/IP protocols!
Only change the link layer (L2) and physical layer (L1) technologies.

```
Application  →  HTTP, SSH, DNS
Transport    →  TCP / UDP
Network      →  IP   ← Internet
Link         → ***** Wifi
Physical     → ***** Radio
```

WiFi (Wireless Fidelity)

IEEE 802.11 standards define the link layer and physical layer for wireless local area networks (WLANs), commonly known as WiFi.

WiFi Standards Evolution

Standard	Year	Frequency	Max Speed	Range (Indoor)
802.11b	1999	2.4 GHz	11 Mbps	~35m
802.11a	1999	5 GHz	54 Mbps	~25m
802.11g	2003	2.4 GHz	54 Mbps	~38m
802.11n (WiFi 4)	2009	2.4/5 GHz	600 Mbps	~50m
802.11ac (WiFi 5)	2014	5 GHz	6.9 Gbps	~40m
802.11ax (WiFi 6)	2019	2.4/5 GHz	9.6 Gbps	~45m
802.11ax (WiFi 6E)	2020	6 GHz	9.6 Gbps	~40m
802.11be (WiFi 7)	2024	2.4/5/6 GHz	46 Gbps	~45m

Most common today:

- **Home routers / clients:**
  Wi-Fi 5 (802.11ac) or Wi-Fi 6 (802.11ax)
- **IoT devices (general):**
  Wi-Fi 4 (802.11n) — cheaper, lower power, 2.4 GHz friendly
- **ESP8266:**
  Wi-Fi 4 (802.11b/g/n), 2.4 GHz only, no Bluetooth

- **ESP32 (classic / S2 / S3):**
  Wi-Fi 4 (802.11b/g/n) + Bluetooth (Classic + BLE*)
- **ESP32-C3:**
  Wi-Fi 4 (802.11b/g/n) + BLE 5 (no Classic BT)
- **ESP32-C6:**
  Wi-Fi 6 (802.11ax, 2.4 GHz only)
    - Bluetooth LE 5
    - IEEE 802.15.4 (Thread / Zigbee / Matter)

ESP32C6 WiFi

We can use ESP32C6 Wifi for the following:

- Web Server / REST API
    - Any device can access it via browser or HTTP client
- HTTP/HTTPS Client
    - Connect to external APIs (weather, news, etc.)
- MQTT Client
    - Publish sensor data to cloud
- SoftAP
    - Create your own WiFi network for direct device access

ESP32C6 Usage Pattern

1. Web Server + MQTT Client

```
Browser → ESP32 (Web Server)
ESP32 → Cloud MQTT Broker
```

2. REST API Server + HTTP Client

```
Mobile App → ESP32 API
ESP32 → Weather API
```

3. SoftAP Server + STA Client

```
Mobile App → ESP32 SoftAP
ESP32 → Home Router (STA)
```

The magic is the virtual intefarce!

```
    Wi–Fi Radio      ← Physical NIC (1)

AP Interface              STA Interface
(SoftAP)                  (Station)
```

WiFi Security Standards

Evolution of Security

```
┌──────────────────────────────────────────────────┐
│ WEP (Wired Equivalent Privacy) - 1997            │
│ ✗ BROKEN - Do NOT use! Can be cracked in minutes │
│ • 64/128-bit keys                                │
│ • RC4 encryption (weak)                          │
└──────────────────────────────────────────────────┘

┌──────────────────────────────────────────────────┐
│ WPA (WiFi Protected Access) - 2003               │
│ • WEAK - Avoid if possiblei (use WPA2 instead)   │
│ • Better than WEP but still vulnerable           │
└──────────────────────────────────────────────────┘
```

```
WPA2 (2004) — CURRENT MINIMUM
✓ Strong security if configured properly
• AES encryption (128-bit)
• Pre-shared key (PSK) or Enterprise (802.1X)
• Still vulnerable to KRACK attack (patched)
```

```
WPA3 (2018) — BEST (if available)
✓ Enhanced security
• 192-bit encryption
• Protection against brute-force attacks
• Forward secrecy (past traffic stays secure)
• SAE (Simultaneous Authentication of Equals)
```

How WiFi Encryption Works

WPA2 Connection Process (4-Way Handshake)

```
Step 1: Device sends connection request
Client → AP: "I want to connect to 'MyWiFi'"

Step 2: AP challenges device
AP → Client: "Prove you know the password" (ANonce)

Step 3: Device proves knowledge
Client → AP: "Here's my proof" (SNonce + MIC)
             Uses password to create Pairwise Transient Key (PTK)

Step 4: AP verifies and confirms
AP → Client: "Verified! Installing keys now" (GTK)
```

The core idea

- The client should know the password in advance.
- Password is never sent over the air!
- Instead, the password is used to derive encryption keys (PTK and GTK).
- The AP computes the same keys based on the password and nonces.
- Each session has unique keys
- Even with same password, past sessions can't be decrypted

For IoT Devices (ESP8266/ESP32)

- Store credentials securely (not in plain text)
- Use HTTPS for web communication
- Implement OTA (Over-The-Air) updates
- Add authentication to web interfaces
- Validate all user inputs
- Use unique passwords per device
- Implement watchdog timer

WiFi vs Ethernet vs Simple Wireless

Metric	Wi-Fi	Ethernet	Bluetooth
Range	30–50 m	100 m / segment	10–100 m
Speed	50–1000 Mbps	100–1000 Mbps	1–3 Mbps
Latency	10–30 ms	1–5 ms	20–100 ms
Security	WPA2 / WPA3	Physical	Pairing / Encryption
Power	High (100–500 mA)	N/A	Low (10–50 mA)
Internet	Yes	Yes	No (needs gateway)
Complexity	High	Low	Medium
Cost	Medium	Low	Low
Interference	High (Wi-Fi, BT, microwave)	None	Medium
Use Case	Internet access	Wired network	Peripherals

When to Use Wifi/EthernetKEach Technology

WiFi

✓ Use when:

- Need quick and simple internet connectivity
- Device mobility required
- Multiple devices (phone, laptop, tablet)
- Non-critical applications (home use, IoT)

✕ Avoid when:

- Battery life critical (< 1 week)
- Ultra-low latency needed (< 5ms)
- Range > 50 meters without repeaters
- Harsh RF environment (many interferers)

Remeber Wifi can be easily hacked and interfered with.

Ethernet (Wired)

✓ Use when:

- Stationary device (desktop, TV, server)
- Reliability critical
- Low latency required (gaming, VoIP)
- High security needs
- Professional and critical applications

✕ Avoid when:

- Mobility required
- Cable installation impractical
- Temporary deployment
- Cost-sensitive (cabling, labor)

Remember wiring is expenisve, but offers best performance and security!